

CROP YIELD PREDICTOR

1. Introduction

A lot of people in India are farmers. They make decisions based on what they think will work, not on data. We want to change this. We made a model that uses machine learning to guess crop yield. The model uses things like rainfall, temperature and soil quality to make predictions.

The project is about farming in India. We used numbers that're like the real conditions in Madhya Pradesh. The model guesses crop yield in quintals per hectare. This is a unit that farmers already use.

2. Problem Statement

We want to answer a question: can we guess crop yield before the harvest using basic farming data? We also want to know which things matter most for yield like rainfall or fertilizer. We want to know which model works best.. We want to make sure the predictions are useful for farmers, not just data scientists.

If we do not guess crop yield correctly farmers might buy much fertilizer or not water their crops enough. A good prediction tool can help with these problems.

3. Dataset

3.1 Overview

We made a dataset with 400 records. Each record is like one farms data for a season. The thing we are trying to guess is Crop Yield in quintals per hectare.

3.2 Features

Here are the things we used to make predictions:

Feature	What It Means
Rainfall	how rain the farm got
Temperature	how hot or cold it was
Soil Fertility	how good the soil is
Fertilizer Used	how fertilizer the farmer used
Pesticide Used	how pesticide the farmer used
Irrigation	how many hours the farmer watered the crops
Previous Year Yield	how much crop yield the farmer got year
Crop Type Code	what kind of crop the farmer is growing
Crop Yield	how much crop yield the farmer will get

3.3 Target Engineering

We made the target variable by combining all the input things with some noise. This makes it more like life.

4. Methodology

4.1 Train-Test Split

We split the dataset into two parts: one for training and one for testing. This helps us make sure the model is working correctly.

4.2 Models Trained

We trained three models: Linear Regression, Decision Tree Regressor and Random Forest Regressor. We chose these models because they are good at guessing numbers.

4.3 Evaluation Metrics

We used three metrics to evaluate the models:

- MAE: how off each prediction is
- RMSE: like MAE but it punishes bigger errors more

- R^2 Score: how well the model explains the data

5. Results

5.1 Model Performance

The Random Forest Regressor was the model. It had the errors and the highest R^2 score.

Model	MAE	RMSE	R^2 Score
Linear Regression	4.xx	5.xx	0.xx
Decision Tree Regressor	3.xx	4.xx	0.xx
Random Forest Regressor ☒	2.xx	3.xx	0.xx

5.2 Sample Prediction

We tested the model with a sample farm. The farm had conditions. The model guessed a crop yield.

6. Discussion

6.1 What the Model Gets Right

The Random Forest model is good at capturing how the different things interact. For example it knows that rainfall and good soil make for better crop yield.

6.2 Limitations

- The dataset is fake not real.
- The model does not learn from trends over time.
- The model does not account for problems, like droughts.

6.3 What Could Be Improved

- We could use data from farms.
- We could add a weather forecast to make predictions in time.
- We could use a way of encoding the crop type.

7. Key Decisions We Made

7.1 Why We Chose Random Forest as the Final Model

We trained three models and compared them. We chose Random Forest because it gave the best results. It had the lowest errors and the highest R^2 score. Linear Regression was too simple and Decision Tree was okay but not as good. Random Forest uses many trees and combines their answers. This makes it more reliable.

7.2 Why 400 Records

We used 400 records because it is a good number for testing. It is not too small and not too big. It gives the model enough data to learn from and still leaves enough to test on.

7.3 Why 75/25 Split

We used 75 percent of the data for training and 25 percent for testing. This is a common way to do it. It gives the model enough data to learn and still leaves enough to check if it is working correctly.

7.4 Why We Used Synthetic Data

We could not get real farm data easily. So we made our own data using numbers that are like the real conditions in Madhya Pradesh. This helped us build and test the model even without real data.

8. Challenges We Faced

- Getting real farm data was hard. We could not find a dataset that had all the things we needed. So we had to make our own data.
- Choosing the right model took time. We had to train all three models and compare them before we knew which one was best.
- Understanding the evaluation metrics was not easy at first. We had to learn what MAE, RMSE and R^2 Score mean and how to use them to compare models.
- Making the prediction useful was a challenge. It is not enough to just give a number. We also had to tell the farmer if the yield is good or not and what to do about it.
- The model does not know about things like crop diseases or sudden weather changes. This made it hard to make the predictions fully realistic.

9. What We Learned

- We learned that machine learning can be useful for farming problems. It is not just for tech companies. It can help real people in real situations.
- We learned that Random Forest is a strong model for predicting numbers. It works well even when the data has many different things affecting the result.
- We learned that data quality matters a lot. Even with a good model, if the data is not realistic the predictions will not be useful.
- We learned how to evaluate models properly. Not just by looking at one number but by comparing MAE, RMSE and R² Score together.
- We learned that a prediction is more useful when it comes with a suggestion. Just saying the yield will be 34 quintals is not enough. Telling the farmer what to do next is what makes it helpful.
- We learned that there is a big gap between a model that works on a computer and a tool that a farmer can actually use. Bridging that gap is the next big challenge.

10. Conclusion

This project shows that machine learning can help farmers make decisions. The Random Forest Regressor was the model. It guessed crop yield correctly. Identified the most important things that affect yield.

The model can help farmers decide what to plant how much to water and when to use fertilizer. This can make farming better.

11. Technology Stack

We used Python, pandas, numpy, scikit-learn, matplotlib, seaborn and joblib. These tools helped us make the model and evaluate it.

Tool	Used For
Python	Main programming language
pandas	Data handling
numpy	Numbers and data generation
scikit-learn	Model training and evaluation
matplotlib	Plotting
seaborn	Heatmap visualization
joblib	Saving and loading the model